Google Summer of Code 2026 – Project Proposal

AI-Accelerated Reconstruction for the ATLAS Tile Calorimeter at the HL-LHC

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Abstract

Signal reconstruction in calorimeters requires precise estimation of pulse amplitude and timing from overlapping electronic signals. At the High-Luminosity LHC, the ATLAS Tile Calorimeter will face unprecedented pile-up conditions where dozens of simultaneous proton collisions corrupt individual pulse shapes and degrade reconstruction quality. Traditional algorithms like Optimal Filtering rely on fixed pulse templates and struggle in these regimes. This GSoC project proposes developing compact deep-learning models such as convolutional or recurrent neural networks to reconstruct TileCal signals with higher accuracy and robustness. I will design, train, and benchmark models against baseline algorithms, exploring optimization techniques like quantization, knowledge distillation and pruning to meet the strict latency and resource constraints required for FPGA deployment. The best-performing optimized model will be integrated into the ATLAS offline reconstruction framework, Athena, as a drop-in replacement for the existing amplitude estimation step, using the ONNX export format and the `lwtnn` or ONNXRuntime inference backend. The expected outcome is a reproducible ML reconstruction pipeline with quantitative performance comparisons under HL-LHC-like conditions, alongside a validated, open-source Athena algorithm prototype contributing directly to the TileCal readout upgrade R&D

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2 About Me

2.1 Experience in Machine Learning and Software Development

I am a sophomore majoring in Computer Science and Engineering with a strong focus on artificial intelligence, machine learning, and deep learning. My research and software development experience ranges from building foundational models from scratch to applying machine learning techniques to complex scientific problems.

Recently, I completed the GSoC Phase 1 evaluation project for CERN-ATLAS, where I successfully built and analyzed a ridge regression model for Tile Calorimeter signal reconstruction.

On the development side, I mainly work with Python (especially PyTorch) and Rust, and I've also built a few full-stack projects using React and TypeScript. I like working on projects that combine systems thinking with machine learning, especially when they involve solving interesting or unconventional problems.

2.2 My Motivation

As a developer fascinated by the fundamental laws of nature, I have always been driven to explore the intersection of physics and machine learning. Therefore, this project proposal immediately caught my attention, as it aligns perfectly with my ambition to build computationally efficient models for scientific discovery.

In the upcoming era of the High-Luminosity LHC (HL-LHC), the sheer volume and complexity of data, specifically the extreme pile-up conditions, present a formidable computational bottleneck. It is crucial that we evolve our analytical methods to ensure we can accurately capture and process these physics events. This project directly addresses this challenge by aiming to transition traditional Tile Calorimeter signal reconstruction into a robust, deep-learning-accelerated pipeline.

Having already engaged deeply with the Tile Calorimeter data and baseline reconstruction algorithms during the evaluation phase, I am eager to expand this work significantly. Through this project, I hope to gain a more profound understanding of how advanced neural architectures can be optimized for the demanding environments of high-energy physics. I look forward to the challenge of scaling these models to handle intense pile-up scenarios without sacrificing accuracy or computational efficiency.

Furthermore, my continued exploration of Physics-Informed Neural Networks (PINNs) and quantum computing reinforces my belief that the future of particle physics relies heavily on innovative computational paradigms. By developing these accelerated pipelines, we can directly

enhance the discovery potential of the ATLAS experiment. This is a thrilling frontier, and I am eager to contribute my skills and knowledge towards this goal.

2.3 Why CERN-HSF?

I like pure science, deep learning, physics-informed machine learning, research, and open source and CERN-HSF sits at the heart of all of these. It is the only organisation whose work truly aligns with everything I am passionate about, which is why I am applying exclusively to CERN-HSF for GSoC 2026 and do not intend to apply anywhere else. I have worked on deep learning models and explored hardware-aware inference optimization, and I want to apply those skills to a problem that genuinely matters. The TileCal reconstruction challenge is not just an interesting technical problem, it is a real bottleneck for the future of the ATLAS detector, and contributing to its solution feels meaningful in a way that few other projects do. CERN-HSF's commitment to open science and collaborative development aligns deeply with how I believe research should be done. My long-term goal is to pursue a PhD at the intersection of machine learning and experimental physics, and this project would give me exactly the kind of rigorous, real-world research experience I am looking for.

2.4 Relevant Projects

- **CERN-ATLAS Signal Reconstruction (GSoC Phase 1):** Developed a ridge regression model for Tile Calorimeter signal reconstruction, achieving an R^2 of 0.9906 and a mean relative error of 0.41 on held-out test data. Analyzed dataset structure, implemented a full PyTorch training pipeline using AdamW with L2 regularization, and generated diagnostic plots across the full energy spectrum.
- **Verbatex:** Currently developing a Transformer-based model that converts natural language into LaTeX mathematical expressions, aimed at streamlining note-taking and technical documentation.
- **Physics-Informed Neural Networks (PINNs):** Designed and implemented a PINN with a custom loss function to model steady-state fluid flow, integrating physical constraints directly into the learning process.
- **Gravitational Lensing Detection:** Implemented a Convolutional Neural Network (CNN) to identify complex gravitational lensing events in astronomical datasets.
- **Neural Networks from First Principles:** Implemented a foundational neural network from scratch in Rust and a Transformer architecture from scratch in PyTorch to deeply understand learning dynamics, backpropagation, and sequence modeling.

3 Project Background

3.1 Introduction: The ATLAS Tile Calorimeter

The ATLAS Tile Calorimeter (TileCal) is the central hadronic calorimeter of the ATLAS experiment at the Large Hadron Collider (LHC) at CERN. It serves as one of the principal instruments for measuring the energy deposited by hadronic jets, contributing to the reconstruction of missing transverse energy, and assisting in muon identification and triggering [2]. TileCal is a sampling calorimeter that uses steel plates as absorber material and plastic scintillating tiles as the active medium. Charged particles traversing the scintillators produce light, which is guided via wavelength-shifting (WLS) optical fibres to photomultiplier tubes (PMTs) for readout.

The detector consists of a central Long Barrel (LBA and LBC) and two Extended Barrels (EBA and EBC), each divided into 64 modules, with up to 45 PMTs per module amounting to approximately 5,182 readout cells and 9,852 channels in total [1]. Each cell is read out redundantly by two PMTs. The readout architecture segments the calorimeter into pseudo-projective cells organised in depth layers, enabling three-dimensional hadronic energy flow reconstruction.

Signals from each PMT are digitized by 12-bit Analog-to-Digital Converters (ADCs) at the LHC bunch-crossing frequency of 40 MHz, producing one sample every 25 ns. A window of seven consecutive samples centred on the in-time bunch crossing is used to characterise the pulse. Three primary parameters fully describe any single TileCal pulse: the *pedestal* (baseline ADC count in the absence of signal), the *amplitude* (proportional to the deposited energy), and the *phase* (time shift of the pulse peak relative to the nominal bunch crossing). The reconstruction task is to extract the amplitude and phase from this sampled window with maximum precision and minimum latency [7].

3.2 The HL-LHC Challenge: Why Existing Methods Are No Longer Sufficient

3.2.1 Unprecedented Pile-up at the HL-LHC

The High-Luminosity LHC (HL-LHC) upgrade, scheduled to begin around 2029, will increase the instantaneous luminosity by roughly a factor of five to seven relative to Run 3 conditions, targeting a peak luminosity of $\mathcal{L} = 7.5 \times 10^{34}$ cm^{-2}s^{-1} [6]. This corresponds to an average number of simultaneous proton-proton interactions per bunch crossing of $\langle\mu\rangle \approx 200$, compared to roughly 50–60 in Run 3. At this occupancy, the 25 ns bunch spacing means that TileCal pulses from adjacent bunch crossings overlap extensively within the same 7-sample readout window a phenomenon known as *out-of-time pile-up*.

The TileCal pulse shape is wider in time than the LHC bunch spacing. In high-occupancy cells such as those in the E layer, adjacent collisions are therefore observed within the same readout window, causing the resultant signal to be a superposition of multiple overlapping pulses. This deformation severely compromises amplitude estimation if the algorithm assumes a single clean pulse. Reconstruction performance under Run 3 conditions and future perspectives are discussed in [5, 8].

3.2.2 Limitations of the Classical Optimal Filtering Algorithm

The classical reconstruction algorithm currently deployed in TileCal is the *Optimal Filtering* (OF) method [1, 2, 4]. OF2, the version in operational use, estimates the amplitude $\hat{A}$ as a weighted linear combination of the $N = 7$ ADC samples:

$$\hat{A} = \sum_{i=1}^{N} s_i w_i \tag{1}$$

where s_i are the ADC samples and w_i are weights derived offline by minimising the variance of the estimator under constraints on phase immunity and baseline rejection [4]. The weight computation relies on two key assumptions: (1) the signal follows a *deterministic, known pulse shape*, and (2) the noise is *Gaussian and uncorrelated* across samples.

Both assumptions break down at HL-LHC pile-up levels. Out-of-time pile-up introduces non-Gaussian, asymmetric noise components whose statistics cannot be captured by the OF noise covariance matrix. The fixed pulse template becomes unreliable when overlapping pulses distort the effective shape seen in the readout window. As a result, the OF algorithm produces systematically biased amplitude estimates, particularly for low-energy signals where the signal-to-noise ratio is already unfavourable. The algorithm also generates a significant negative tail in the energy distribution, corresponding to bunch crossings where an out-of-time pulse partially

cancels the in-time signal [2]. The implementation and performance of signal reconstruction in TileCal using the OF method have been extensively characterised in [7].

3.2.3 Real-Time Constraints for the Phase-II Readout Upgrade

The Phase-II upgrade introduces a new readout architecture in which TileCal data are streamed from on-detector electronics to off-detector Compact Processing Modules (CPMs) at 40 MHz [3]. Each CPM houses Xilinx Kintex UltraScale KU115 FPGAs that must perform per-bunch-crossing energy and time reconstruction on the incoming data stream and forward trigger primitives to the ATLAS Level-0 trigger within a fixed latency budget [1]. The energy reconstruction step is allocated approximately 225 ns (9 bunch crossings) of the total 325 ns firmware pipeline budget.

This constraint imposes severe requirements on any replacement algorithm:

- **Strict linearity or near-linear inference:** Operations must be implementable as fixed-point multiply-accumulate (MAC) arrays. Deep non-linear networks require careful quantization and architecture choices to meet timing.
- **Minimal parameter count:** The CPM must serve up to 154 channels simultaneously (77 per gain for two gain channels), so per-channel model footprints must remain small targeting $N_{\text{params}} \lesssim 150$ for FPGA resource efficiency [1].
- **Reproducibility and stability:** Inference results must be deterministic and bit-exact across firmware versions, necessitating integer or fixed-point arithmetic.

The design and architecture of the CPM for the ATLAS TileCal Phase-II upgrade are described in detail in [3]. The full scope of the TileCal Phase-II hardware upgrade is documented in [6].

3.2.4 The Dual-Gain Readout and Dynamic Range

TileCal employs a dual-gain readout system to maintain energy resolution across a wide dynamic range. The High-Gain (HG) channel provides sensitivity to small energy deposits below ∼10 GeV but saturates for large signals. The Low-Gain (LG) channel covers high-energy deposits without saturation but has coarser resolution at low energies. Both channels use 12-bit ADCs, with a gain ratio of approximately 40. This duality means that a complete reconstruction pipeline must handle both channels, either through two separate models or a single model operating on a combined gain-switched input.

3.3 Phase 1 Evaluation: Linear Reconstruction Results

As part of the GSoC 2026 Phase 1 evaluation, I developed and benchmarked two fully linear reconstruction models for the LG channel of the ATLAS TileCal simulation dataset. Both share a single `nn.Linear(7, 1)` architecture, a data-driven weighted sum of the form $\hat{y} = \mathbf{w}^\top \mathbf{x} + b$, making them directly FPGA-compatible. The dataset comprised approximately 534K training, 115K validation, and 115K test events drawn from PyTorch shard files with input shape $[2048, 2, 7]$. The two models differ only in regularization: a baseline trained with Adam on MSE, and a Ridge Regression variant trained with AdamW ($\lambda = 10^{-3}$) to suppress large weights on noisy out-of-time pulse wings.

Table 1: Test-set performance on the LG channel: Baseline vs. Ridge Regression.

Metric	**Linear Regression**	**Ridge Regression**
MSE	0.3240	**0.0111**
MAE	0.0632	**0.0302**
R^2 Score	0.7263	**0.9906**
Mean Relative Error	2.4496	**0.4122**
RMS Relative Error	30.3146	**10.0534**

The Ridge model achieves $R^2 = 0.9906$ with an MSE reduction of 29× over the baseline, performing well across the mid-to-high energy range. The elevated RMS relative error (10.05 vs. mean of 0.41) is driven by low-energy events where pile-up completely obscures the true signal, a known limitation shared with MLP and CNN models on the same dataset [1, 2]. This hard floor for linear methods directly motivates the move to compact non-linear architectures in the main GSoC project.

3.4 Comparison with Existing Approaches and the Case for Deep Learning

3.4.1 Optimal Filtering (Current Baseline)

The Optimal Filtering algorithm [2, 4] is computationally minimal 7 multiplications and 6 additions per channel per bunch crossing and has been the operational standard in TileCal since initial LHC runs [7]. Its weights are derived analytically from the known pulse shape and noise covariance matrix, making it interpretable and stable. However, OF assumes a single-pulse, Gaussian-noise model that is fundamentally violated under $\langle\mu\rangle = 200$ pile-up, producing large systematic biases particularly in the negative energy tail from out-of-time contributions. Energy reconstruction performance during Run 2 using this method is discussed in [5].

3.4.2 Wiener Filter

The Wiener filter, investigated by Oliveira et al. [2], extends the OF framework by deriving weights from the full statistical properties of both signal and noise through the cross-correlation vector $\mathbf{p}$ and autocorrelation matrix $\mathbf{R}$:

$$\mathbf{w} = \mathbf{R}^{-1}\mathbf{p} \tag{2}$$

Applied to EBA E4 cells at $\langle\mu\rangle \approx 90$ during the 2018 special high-μ run, the Wiener filter demonstrated a marked reduction in the negative energy tail compared to OF2, with an energy distribution RMS reduced from 498 MeV to 325 MeV. The bias constant term introduced in the filter design makes it insensitive to luminosity variation. Nevertheless, the Wiener filter remains a linear method and therefore cannot absorb the higher-order (non-Gaussian) statistics of pile-up noise. Its performance improvement saturates as $\langle\mu\rangle$ grows toward 200.

3.4.3 Data-Driven Linear Models (Phase 1 Result)

The ridge regression model developed during Phase 1 achieves $R^2 = 0.9906$ on the LG channel under simulated $\langle\mu\rangle = 200$ conditions, demonstrating that data-driven weight optimization substantially outperforms analytic OF weight derivation when training data is available. The regularized linear model is fully FPGA-compatible and matches the latency and resource footprint of the classical OF. However, the low-energy pile-up floor remains a hard limit for linear methods.

3.4.4 Compact MLP and CNN Models (State of the Art)

The current baseline for ML-driven reconstruction comes from recent work by Curcio et al. [1], who evaluated highly constrained networks ($N_{\text{params}} \approx 150$) on the exact same $\langle\mu\rangle = 200$ dataset. Using a 9-sample input window, they tested a standard three-layer Multilayer Perceptron (hidden dimensions 9 and 4, PReLU activations), which yielded an average absolute error of 99.76 ADC counts on the High-Gain channel and 10.75 on the Low-Gain. However, they found that switching to a 1D convolutional approach (two layers with kernel size 3) allowed the model to exploit local temporal correlations between adjacent samples. This CNN architecture significantly outperformed the MLP, dropping the error to 72.12 ADC counts (HG) and 8.36 (LG).

The CNN consistently outperforms the MLP, particularly on the HG channel where it eliminates the step artefact at $(E_{\text{pred}} - E_{\text{true}})/E_{\text{true}} = -1$ that persists in the MLP. The authors attribute this advantage to the CNN's capacity to exploit local temporal correlations between adjacent bunch crossings within the input window. Both models are trained with a hybrid loss $\mathcal{L} = 0.5 \cdot \text{MAE} + 0.5 \cdot \text{RMSE}$ and use 9-sample input windows combining HG and LG gain-switched samples.

Table 2: Summary comparison of reconstruction approaches for ATLAS TileCal under HL-LHC pile-up conditions.

Method	**Linear?**	**Pile-up Robust**	**FPGA Compatible**	σ_{avg} **(LG)**
Optimal Filtering (OF2)	Yes	No	Yes	—
Wiener Filter	Yes	Partial	Yes	RMS 325 MeV
Ridge Regression (Phase 1)	Yes	Partial	Yes	MAE 0.030 (norm.)
MLP ($N_{\text{params}} \approx 148$)	No	Yes	Possible	10.75 ADC cts
CNN ($N_{\text{params}} \approx 147$)	No	**Yes**	Possible	**8.36 ADC cts**

3.4.5 Rationale for the Proposed Approach

The comparison above indicates a clear trend. Linear methods whether analytic (OF, Wiener) or data-driven (ridge regression) approach their fundamental performance ceiling under $\langle\mu\rangle = 200$ pile-up. The non-linear, non-Gaussian contributions from overlapping pulses require models capable of learning higher-order statistical structure. At the same time, the FPGA deployment constraint rules out large, computationally dense networks. The CNN at $\sim$147 parameters represents the current state of the art, but it has not been systematically explored for further gains from model optimization techniques such as quantization-aware training, knowledge distillation from larger teacher models, or pruning all of which could simultaneously improve reconstruction accuracy and reduce FPGA resource usage.

This GSoC project picks up precisely where the existing literature leaves off: taking the compact CNN as a starting point, extending the architecture search space, and rigorously benchmarking optimized models against both classical baselines and the Phase 1 linear results, with a clear path toward hardware-compatible deployment.

4 Proposed Project & Technical Approach

4.1 Deep Learning Architectures for Reconstruction

The Phase 1 results and the existing literature jointly establish that the low-energy pile-up regime is a hard floor for linear methods: no matter how well-tuned the weights, a single weighted sum cannot disentangle overlapping non-Gaussian pulse contributions. The transition to non-linear models is therefore not a matter of preference but of physical necessity. The

central design constraint remains unchanged: any proposed architecture must be compact enough ($N_{\text{params}} \lesssim 150$–$200$) to fit alongside 154 simultaneous channel instances on a Xilinx Kintex UltraScale KU115 FPGA within the 225 ns energy reconstruction budget [3].

4.1.1 1D Convolutional Neural Networks

The 1D CNN is the primary candidate architecture and the current state of the art for this problem [1]. Its inductive bias, local receptive fields over the time axis, maps directly onto the physics: each convolutional kernel learns to detect characteristic local pulse shape features (rising edge, peak, falling tail) and their distortions under pile-up, while translational equivariance along the sample dimension reflects the fact that the pulse shape is the same regardless of which samples it falls on. The existing CNN baseline uses two Conv1d layers (6 and 4 feature maps, kernel size 3, same padding) with PReLU activations and a final dense layer, totalling 147 parameters on a 9-sample input.

In this project I plan to extend the CNN search space along three axes: (1) varying the number of convolutional layers (1–3) and feature map counts to find the optimal capacity-to-parameter trade-off; (2) exploring dilated convolutions with dilation rates of 2 and 3 to capture longer-range inter-bunch-crossing correlations without increasing the kernel size or parameter count; and (3) evaluating depthwise separable convolutions as a parameter-efficient alternative to standard convolutions, which decompose the channel-mixing and spatial-filtering operations to reduce multiply-accumulate counts on the FPGA.

4.1.2 Compact Multilayer Perceptrons

The compact MLP serves as an important ablation point. The existing baseline (three linear layers with PReLU, 148 parameters) already demonstrates that non-linear models outperform the linear OF even at the same parameter budget [1]. I will explore whether architectural modifications such as residual connections between layers, layer normalization applied before each PReLU, and different hidden dimension schedules improve MLP performance without exceeding the parameter budget. The MLP is also the most FPGA-friendly non-linear architecture because its datapath reduces to a sequence of matrix-vector multiplications, each implementable as a single DSP-optimized MAC array stage.

4.1.3 Gated Recurrent Units for Sequence Modeling

The 7-sample ADC window is a time-ordered sequence, and the pile-up corruption is causally structured: out-of-time contributions from earlier bunch crossings arrive in samples $n-3$ through $n-1$, while later contributions contaminate samples $n+1$ through $n+3$. Gated Recurrent Units (GRUs) are well-suited to exploit this causal structure because the gating mechanism learns to selectively retain or discard information from earlier time steps when processing the current sample. A minimal single-layer GRU with a hidden state of dimension 8–16 introduces roughly 100–250 parameters and can be unrolled at compile time into a fixed-depth combinational circuit on the FPGA, avoiding the latency penalty of sequential recurrence at inference time. I will benchmark a compact GRU against the CNN baseline to determine whether sequence modeling provides additional gains over local convolutional feature extraction for this specific pulse reconstruction task.

4.1.4 Training Protocol and Loss Function

All models will be trained on the same dataset splits used in Phase 1 (261 training shards, 56 validation, 56 test, approximately 534K / 115K / 115K events). A comparable train/validation/test partitioning will be adopted, with adjustments made based on the final dataset provided during

the project to ensure consistency and fair evaluation. Following Curcio et al. [1], I will use the hybrid loss

$$\mathcal{L} = \alpha \cdot \text{MAE} + \beta \cdot \text{RMSE}, \quad \alpha = \beta = 0.5 \tag{3}$$

which balances sensitivity to large outlier errors (RMSE term) with robustness to the low-energy pile-up tail (MAE term). I will additionally evaluate a weighted relative error loss that explicitly penalizes reconstruction errors in proportion to the true energy, which may better target performance in the low-energy regime where the pile-up floor currently dominates the RMS relative error metric. All models will use the AdamW optimizer with cosine annealing learning rate scheduling and early stopping on validation MAE.

4.2 Model Optimization & Hardware Constraints

Meeting the FPGA deployment constraints requires not only a compact architecture but also careful numerical representation and post-training compression. The target hardware, the Xilinx Kintex UltraScale KU115, performs fixed-point arithmetic natively, and the 225 ns latency budget admits only a small number of pipeline stages [3]. This subsection details the three-stage optimization strategy I will apply to each candidate architecture after initial floating-point training.

4.2.1 Quantization-Aware Training

Straightforward post-training quantization (PTQ) of weights and activations to 8-bit integers typically degrades reconstruction accuracy significantly for small models, because each individual weight carries a proportionally large fraction of the model's representational capacity. Quantization-Aware Training (QAT) addresses this by inserting fake quantization operators into the forward pass during training, so that the model learns weight values that are robust to the discretization that will occur at deployment. I will implement QAT using PyTorch's `torch.ao.quantization` framework, simulating fixed-point formats of 8-bit and 16-bit for weights and activations. The key hyperparameter is the number of fractional bits, which controls the trade-off between dynamic range and precision. I will perform a grid search over bit-widths $\{6, 8, 12, 16\}$ for both weights and activations, measuring the degradation in R^2 and mean relative error relative to the floating-point baseline. The target is to identify the minimum bit-width that preserves $R^2 \geq 0.990$ on the LG test set.

4.2.2 Weight Pruning

Structured pruning removes entire neurons, filters, or channels from the network, reducing both parameter count and the number of MAC operations at inference time. I will apply iterative magnitude pruning using PyTorch's `torch.nn.utils.prune` module, starting from the best-performing floating-point model and progressively removing the lowest-magnitude weights in increments of 10%, retraining for 20 epochs after each pruning step to allow the remaining weights to compensate. For the CNN architecture, I will apply filter-level pruning on convolutional layers (removing entire output feature maps) rather than unstructured weight-level pruning, because filter removal directly reduces the MAC count per inference pass in a hardware-friendly way. The pruning sweep will target a final sparsity of 30–50% while maintaining reconstruction performance within 5% of the unpruned baseline.

4.2.3 Knowledge Distillation

Knowledge distillation trains a compact *student* model to reproduce the output distribution of a larger, more accurate *teacher* model, transferring representational knowledge that the student

could not learn from hard labels alone. I will train a teacher model with 5–10× more parameters than the deployment target (approximately 750–1500 parameters, still small by ML standards but well beyond the FPGA budget), then use it to supervise a compact student via the distillation loss

$$\mathcal{L}_{\text{distill}} = (1-\gamma)\,\mathcal{L}_{\text{task}}(\hat{y}_s, y) + \gamma\,\text{MSE}(\hat{y}_s, \hat{y}_t) \tag{4}$$

where $\hat{y}_s$ and $\hat{y}_t$ are the student and teacher predictions respectively, y is the ground-truth energy, and $\gamma \in [0, 1]$ is a mixing coefficient optimized on the validation set. Knowledge distillation is particularly well-motivated here because the teacher can learn to smooth over the discontinuous pile-up floor artefact in the low-energy regime, providing softer regression targets that carry more information than the hard true energy labels alone.

4.2.4 Benchmarking and Latency Estimation

All optimized models will be benchmarked against the Phase 1 ridge regression baseline and the Curcio et al. CNN on the common test set using MSE, MAE, R^2, mean relative error, and RMS relative error metrics. To estimate FPGA latency without requiring full firmware synthesis, I will use the `hls4ml` library, which translates PyTorch models to HLS (High-Level Synthesis) C++ and reports estimated latency and resource utilization for Xilinx UltraScale targets. This will allow me to produce a Pareto frontier of reconstruction accuracy versus estimated FPGA latency for all candidate architectures and optimization configurations.

4.3 Integration with the ATLAS Athena Framework

A key deliverable of this project is the integration of the trained reconstruction algorithm into the ATLAS offline software framework, Athena, as a reusable open-source contribution to the CERN-HSF ecosystem. Athena is the C++/Python framework used for all ATLAS simulation, reconstruction, and analysis workflows, and it follows the Gaudi component model in which reconstruction algorithms are implemented as `Algorithm` classes that read from and write to a transient event store.

4.3.1 Architecture of the Integration

The integration will follow the standard TileCal reconstruction chain. The existing TileCal offline reconstruction in Athena passes raw digit objects (`TileDigit`, containing the 7-sample ADC window per channel per bunch crossing) through a sequence of reconstruction algorithms that ultimately produce calibrated cell energy objects (`TileCell`). The ML model will be inserted as a drop-in replacement for the amplitude estimation step, receiving the same `TileDigit` inputs and producing amplitude and phase estimates in the same format as the existing OF algorithm [8].

Concretely, the implementation plan is:

1. **Model export:** Export the trained PyTorch model to ONNX format using `torch.onnx.export`, which provides a hardware-agnostic intermediate representation compatible with multiple inference backends.

2. **Inference Backend Integration:** Integrate the ONNX model using the `lwtnn` (lightweight neural network) library, which is already used within Athena for other ML-based reconstruction algorithms and provides efficient CPU inference with no external runtime dependencies. Alternatively, use the `ONNXRuntime` Athena interface if `lwtnn` does not support the required operators.

3. **Athena Algorithm wrapper:** I will implement a new Athena Algorithm class (e.g., `TileRawChannelMakerML`) in Python. This class will initialize the ONNX model at the job

start, iterate over `TileDigit` collections for each event, execute batch inference, and write the predicted amplitude and phase values to the `TileRawChannel` container in the event store.

4. **Configuration layer:** To ensure the tool is accessible to ATLAS physicists, I will provide a Python configuration fragment utilizing the Athena ComponentAccumulator pattern. This will allow users to seamlessly select the ML algorithm as their primary reconstruction method via standard job options, with easily configurable model paths and inference parameters.

4.3.2 Validation and Testing

To ensure the Athena integration is robust, I will validate the algorithm in two distinct stages. First, I will run the integrated model on the exact simulation samples used for the PyTorch benchmarks. The outputs must match the standalone model down to the expected precision limit, which proves the ONNX export and inference backend have not introduced any numerical bugs.

Second, I will test the algorithm inside the full TileCal offline reconstruction pipeline. By comparing the newly reconstructed cell energies against the baseline OF algorithm on a shared validation dataset, I can confirm that the framework wrapper is not creating unexpected artefacts.

Once validated, all code will be written against the current ATLAS software release. I will include complete documentation detailing the model provenance, training configurations, and usage instructions, and submit the final package as a Merge Request to the ATLAS GitLab repository following CERN-HSF contribution guidelines.

5 Project Goals & Deliverables

The following concrete, measurable outcomes are targeted by the end of the GSoC coding period. Each deliverable is scoped to be achievable within the 350-hour project duration while representing a genuine and lasting contribution to the TileCal Phase-II readout R&D.

1. **Reproducible ML Reconstruction Pipeline.** A fully documented, open-source PyTorch training and evaluation pipeline covering data loading from TileCal shard files, preprocessing, model training with the hybrid MAE+RMSE loss, validation monitoring, and test-set evaluation. The pipeline will be version-controlled and include configuration files specifying all hyperparameters, enabling exact reproduction of all reported results from scratch. This forms the shared foundation on which all subsequent deliverables are built.

2. **Systematic Experiments Across Multiple Model Architectures.** A structured comparison of compact non-linear architectures — 1D CNNs (including dilated and depthwise-separable variants), compact MLPs with residual connections, and GRUs and LSTMs trained and evaluated under identical conditions on the $\langle\mu\rangle = 200$ TileCal simulation dataset. Each architecture will be explored across a range of parameter budgets ($N_{\text{params}} \in [100, 500]$) to map the accuracy-versus-complexity trade-off. Results will be reported for both HG and LG channels using MSE, MAE, R^2, mean relative error, and RMS relative error, enabling direct comparison with the Phase 1 linear baseline and the Curcio et al. CNN benchmark [1].

3. **Trained and Optimized Compact ML Model.** The best-performing architecture from the experiment above, further compressed via quantization-aware training (QAT at 8-bit and 16-bit fixed-point), iterative structured filter pruning (targeting 30–50% sparsity), and knowledge distillation from a larger teacher model. The final model will satisfy the FPGA

deployment constraints ($N_{\text{params}} \lesssim 150$, fixed-point-compatible weights) while achieving the best attainable reconstruction accuracy on the held-out test set. Trained model weights will be released in both PyTorch (`.pt`) and ONNX (`.onnx`) formats for reproducibility and downstream use.

4. **Comparative Benchmark Report.** A quantitative performance report documenting all evaluated models alongside the classical baselines (Optimal Filtering [4], Wiener Filter [2], Phase 1 ridge regression). The report will include 2D distributions of $(E_{\text{pred}} - E_{\text{true}})/E_{\text{true}}$ versus E_{true}, absolute error distributions, predicted-vs-true scatter plots, and estimated FPGA latency and resource utilization from `hls4ml` synthesis reports. This document constitutes the primary scientific output of the project and will be structured for potential inclusion in an ATLAS public note or conference contribution.

5. **Prototype of the Athena Algorithm for Offline Reconstruction.** Development of a working prototype of a `TileRawChannelMakerML` Athena Algorithm that integrates the trained ONNX model into the ATLAS offline reconstruction chain via the `lwtnn` or ONNXRuntime inference backend. The prototype will include a Python ComponentAccumulator configuration fragment, inline documentation, and a validation script confirming numerical agreement between standalone PyTorch inference and Athena-integrated inference on a common test sample. The code will be submitted as a merge request to the ATLAS software repository following CERN-HSF open-source contribution guidelines.

6 Timeline for Deliverables

- **Community Bonding Period:** Engage with mentors Luca Fiorini and Fernando Carriò, review past TileCal reconstruction work and related ATLAS notes [5, 6, 8], and deepen familiarity with the Phase-II readout upgrade architecture [3]. Discuss, investigate, and study related work and literature in greater depth, set up the ATLAS software environment (Athena, `hls4ml`), and confirm access to the TileCal simulation dataset. Throughout this phase, I will also document my progress and write blog posts to share updates and insights with the community.

- **Week 1:** Conduct a focused literature review on compact ML architectures for fast inference (CNNs, MLPs, GRUs), FPGA deployment constraints for HEP triggers, and quantization and pruning techniques. Review the existing TileCal OF firmware pipeline [4,7] to understand the integration target. Finalize the project plan and evaluation metric suite with mentors.

- **Week 2:** Set up the full data pipeline: loading TileCal PyTorch shard files, gain-channel selection, normalization, and sliding-window preprocessing. Reproduce the Phase 1 ridge regression baseline within the new codebase as a sanity check. Establish the shared training configuration (hybrid MAE+RMSE loss, AdamW with cosine annealing, early stopping on validation MAE).

- **Week 3:** Implement and train the compact MLP baseline (matching the Curcio et al. architecture) and the 1D CNN baseline on both HG and LG channels. Produce the first set of diagnostic plots: predicted-vs-true scatter, 2D relative error vs. E_{true}, and absolute error distributions. Compare against the Phase 1 linear baseline.

- **Week 4:** Extend the CNN search space: implement dilated convolutions (dilation rates 2 and 3) and depthwise separable convolutions. Train and evaluate all CNN variants alongside the MLP baseline. Refine architectures based on validation MAE results and identify the top candidates for further optimization.

- **Week 5:** Implement and train compact GRU models (hidden state dimensions 8 and 16, unrolled inference). Benchmark GRUs against the best CNN from Week 4. Explore residual connections and layer normalization in MLP variants. Complete the full architecture comparison table across all models, both channels, and all metrics.
- **Week 6 :** Conduct a comprehensive evaluation of all models developed so far. Prepare a progress report summarizing results, trade-offs, and open questions. Share with mentors for feedback. Publish the first project blog post covering the architecture experiments and key findings from Weeks 1–5.
- **Week 7:** Begin the model optimization phase. Apply quantization-aware training (QAT) to the best-performing architecture, sweeping bit-widths $\{6, 8, 12, 16\}$ for weights and activations. Measure degradation in R^2 and mean relative error relative to the floating-point baseline. Identify the minimum bit-width preserving $R^2 \geq 0.990$.
- **Week 8:** Apply iterative structured filter pruning (10% increments, retrain 20 epochs per step) targeting 30–50% sparsity. Train the knowledge distillation pipeline: fit a larger teacher model ($\sim$750–1500 parameters), then train the compact student with the distillation loss $\mathcal{L}_{\text{distill}}$. Evaluate the combined effect of QAT + pruning + distillation on the final model. Catch up on any pending tasks from earlier weeks.
- **Week 9:** Run `hls4ml` synthesis on the optimized model to obtain estimated FPGA latency and resource utilization. Produce the Pareto frontier of reconstruction accuracy versus latency across all architecture and optimization configurations. Finalize the optimized model and export to ONNX format.
- **Week 10:** Begin the Athena integration. Implement the `TileRawChannelMakerML` Algorithm class, configure the `lwtnn` or ONNXRuntime inference backend, and wire up the `TileDigit` input and `TileRawChannel` output containers. Write the Python ComponentAccumulator configuration fragment. Run the first end-to-end test within Athena. Publish the second blog post covering the optimization results and Athena integration progress.
- **Week 11:** Validate the Athena prototype: verify numerical agreement with standalone PyTorch inference on the common test sample, and compare reconstructed cell energy distributions against the reference OF algorithm. Fix any numerical discrepancies. Complete inline code documentation and write the README covering model provenance, training configuration, and usage instructions.
- **Week 12 — Final Submission:** Finalize all deliverables: clean up and freeze the codebase, submit the Athena Merge Request to the ATLAS GitLab following CERN-HSF guidelines, complete the benchmark report, and publish the final blog post summarizing the full project. Prepare and deliver the final GSoC evaluation presentation. Discuss potential post-GSoC collaborations and next steps with mentors.

7 Commitments

I am flexible with my working hours and can adapt my schedule as needed. This summer, I have no major commitments, and my vacation period will allow me to fully dedicate myself to the project. I can comfortably devote **6–8 hours per day** during this time. From August onwards, my college will resume, but I will manage my academic responsibilities alongside the project without compromising on deliverables. In case of any unavoidable circumstances leading to a temporary reduction in working hours, I will ensure to compensate for the lost time in the

following weeks and will communicate any such situations with my mentors well in advance. I check my emails and messages regularly and my response can be expected within 24 hours. I will be available for regular updates and discussions via commonly used platforms in the **Indian Standard Time Zone (GMT +5:30)**.

8 Post GSoC

As someone who is deeply passionate about the work this organisation is doing at the intersection of machine learning and experimental physics, I cannot imagine simply walking away once the GSoC project is over. I would love to continue contributing to the TileCal reconstruction pipeline, implementing improvements and exploring optimizations that could not be fully covered within the GSoC period, and providing consistent patches and bug fixes along the way. There is so much more to explore whether that is pushing the boundaries of model compression for FPGA deployment, experimenting with new architectures, or improving robustness under varying pile-up conditions. I am also eager to stay engaged with the broader ATLAS and HSF community and take part in development activities beyond the scope of this project. I truly believe that the work being done here has real significance for the future of LHC physics and beyond, and I am excited to remain a part of it long after the summer comes to a close.

9 Conclusion

This proposal presents a technically rigorous and carefully scoped plan to push TileCal signal reconstruction beyond the fundamental limits of linear methods. The Phase 1 evaluation established that data-driven linear models, however well-regularized, cannot overcome the non-Gaussian pile-up floor imposed by $\langle\mu\rangle = 200$ conditions at the HL-LHC. This motivates exploring: compact non-linear architectures, disciplined to fit within the strict parameter and latency budgets of the Xilinx Kintex UltraScale KU115 FPGA [3], optimized through quantization-aware training, structured pruning, and knowledge distillation, and integrated into the ATLAS Athena reconstruction chain as a reusable, open-source contribution. Each deliverable is designed to provide value to the TileCal Phase-II R&D community and collectively constitutes a complete, reproducible ML reconstruction pipeline from raw ADC samples to calibrated energy estimates.

Beyond the technical scope, this work is a direct contribution to the broader mission of open, sustainable, and collaborative scientific computing. The methods developed here, efficient inference under hard hardware constraints, are transferable to any domain requiring real-time signal reconstruction, and their open-source release through CERN-HSF ensures they remain accessible to the community long after the GSoC period ends. Reproducibility, transparency, and collaborative development are not just good engineering practice here they are the foundation on which the next decade of LHC physics will be built.

On a personal level, physics has fascinated me since the earliest days of my education. The realization that the universe is governed by laws precise enough to be written in mathematics, and that experiments like ATLAS exist to probe those laws at their most fundamental, is what drew me toward this work in the first place. For as long as I can remember, I have carried a dream of one day working at CERN – of being part of a community where the deepest questions about the nature of reality are not just discussed but actively pursued. Contributing to that effort through better reconstruction algorithms that help the detector see more clearly through the noise of extreme pile-up feels like a meaningful step toward that dream. I am grateful for the opportunity and fully committed to delivering results that are worthy of the problem.

References

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